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## Revision History

| REV | REVISION NOTES | Date |
|---|---|---|
| X1 | Initial | Jun 10, 2023 |
| A | Final Release | Jul 10, 2023 |
| B | 1.Change 3pin solder pads.<br>2.Update schematic per Ver.A test result. | Sep 15, 2023 |
| B1 | 1.DNP J12&J7.<br>2.Change J10 MPN. | Nov 13, 2023 |
| B2 | 1.Change C21 from 1uf to 4.7uf.<br>2.Change C24 from 0.1uf to 2.2uf.<br>3.Change R82&R192&R193 MPN due to material shortage. | Jan 25, 2024 |

# FRDM-MCXN947

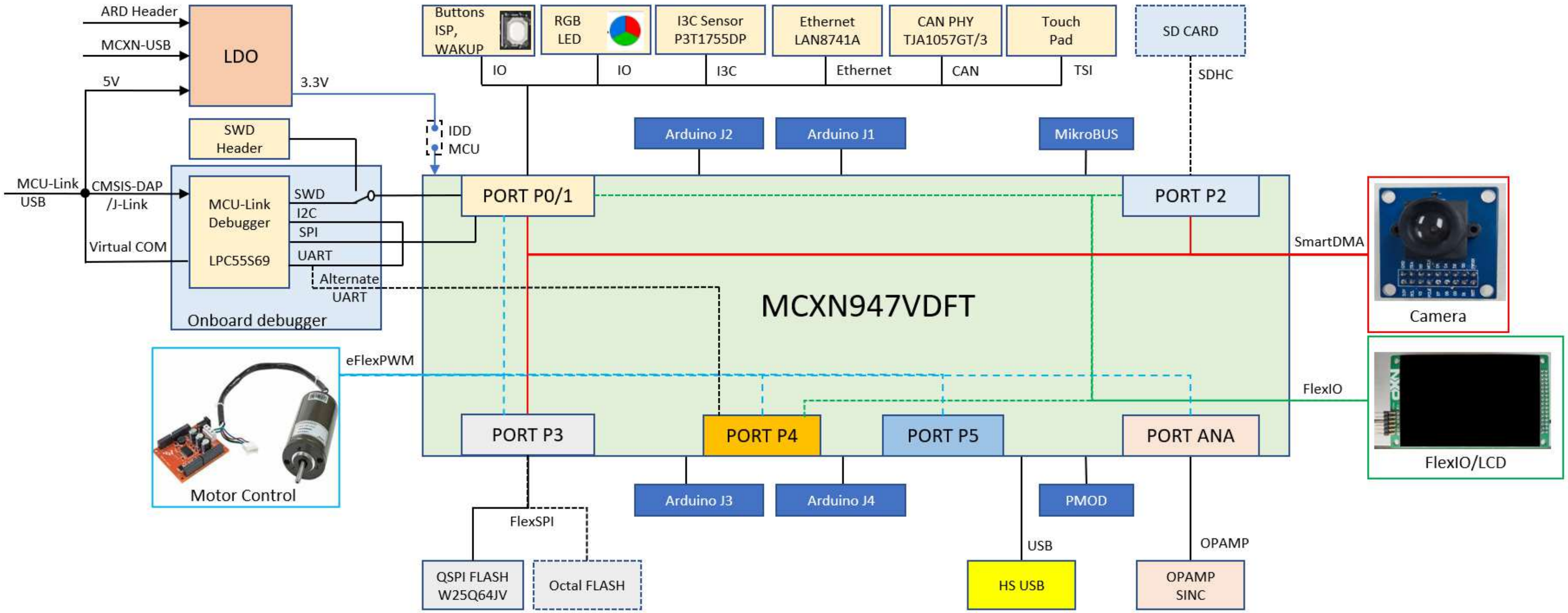

NXP

**Microcontroller Product Group**
6501 William Cannon Drive West
Austin, TX 78735-8598




| | | | |
|---|---|---|---|
| Designer: John Wu | Drawing Title: **FRDM-MCXN947** | | |
| Drawn by: Kate Fan | Page Title: **TITLE PAGE** | | |
| Approved: William Jiang | Size C | Document Number SCH-90818 PDF: SPF-90818 | Rev B2 |
| | Date: Thursday, January 25, 2024 | Sheet 1 of 1 | |

# BLOCK DIAGRAM




Classification: Company Internal/Proprietary

Drawing Title: **FRDM-MCXN947**

Page Title: **BLOCK DIAGRAM**

| Size | Document Number | Rev |
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# SOC PORT0~2


Classification: Company Internal/Proprietary
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Page Title: SOC_PORT0-2

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# SOC PORT3~5


Classification: Company Internal/Proprietary

Drawing Title: **FRDM-MCXN947**

Page Title: **SOC_PORT3-5**

| Size | Document Number | Rev |
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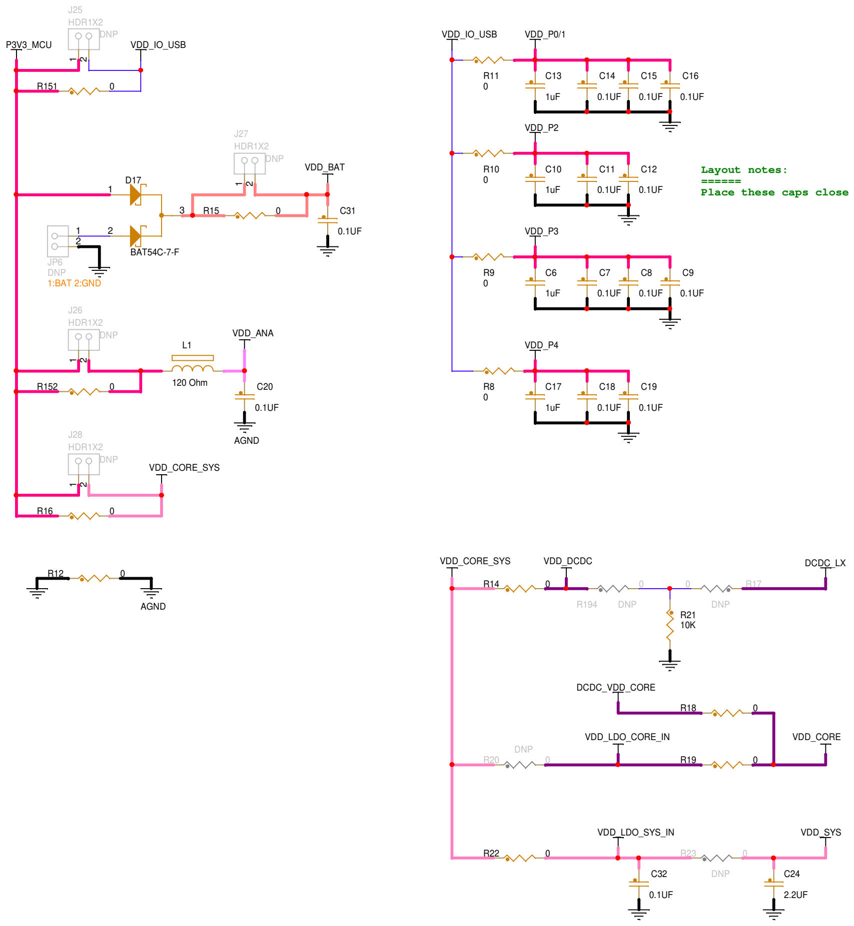
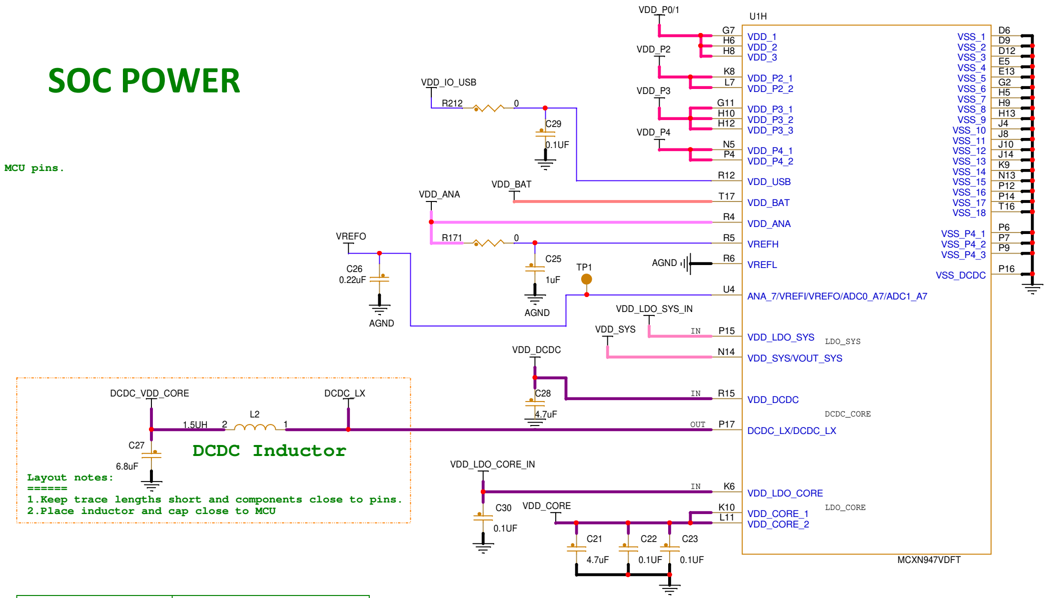
# SOC POWER




Layout notes:
======
Place these caps close to MCU pins.

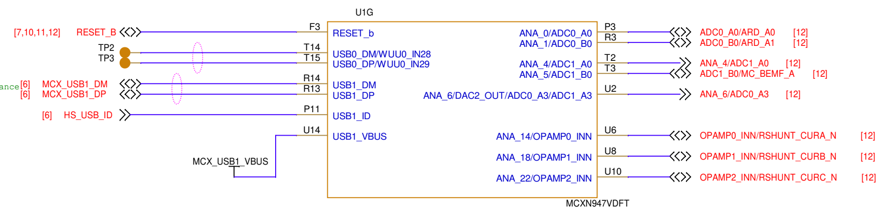



| Mode | Solder Option |
|---|---|
| DCDC_CORE---Enable<br>LDO_CORE---Disable<br>(Default) | Populate R14, R18, R19;<br>DNP R17, R194, R20. |
| DCDC_CORE---Disable<br>LDO_CORE---Enable | Populate R17, R194, R20;<br>DNP R14, R18, R19. |

DNP R23(default): Enable internal LDO_SYS.
Solder R23: Disable internal LDO_SYS.

# SOC USB & Analog Signals

Layout Note:
======
90ohms differential impedance




<Core Design>

| | |
|---|---|
| Classification: | Company Internal/Proprietary |
| Drawing Title: | FRDM-MCXN947 |
| Page Title: | SOC_PWR |
| Size C | Document Number SCH-90818 PDF: SPF-90818 Rev B2 |
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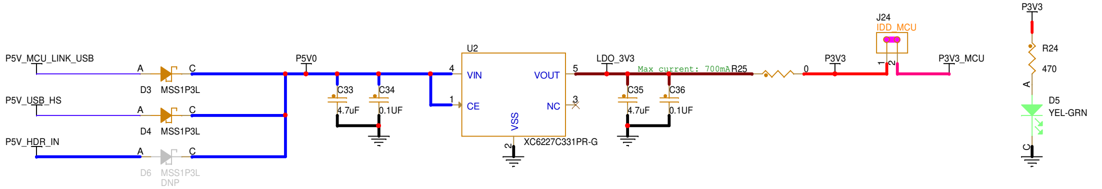
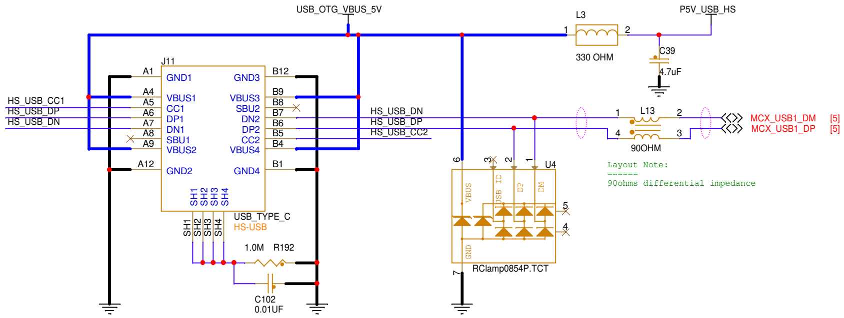
# SYSTEM POWER

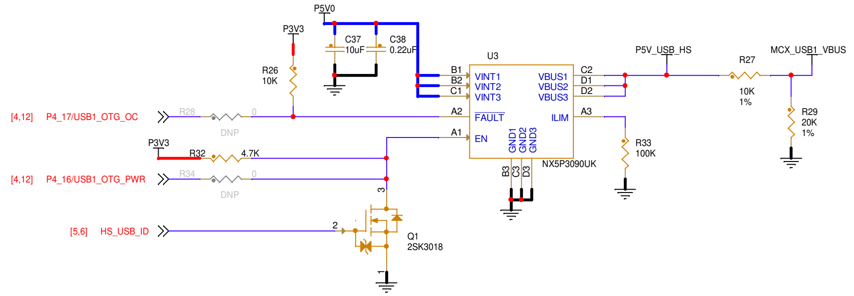



# USB1_HIGH SPEED

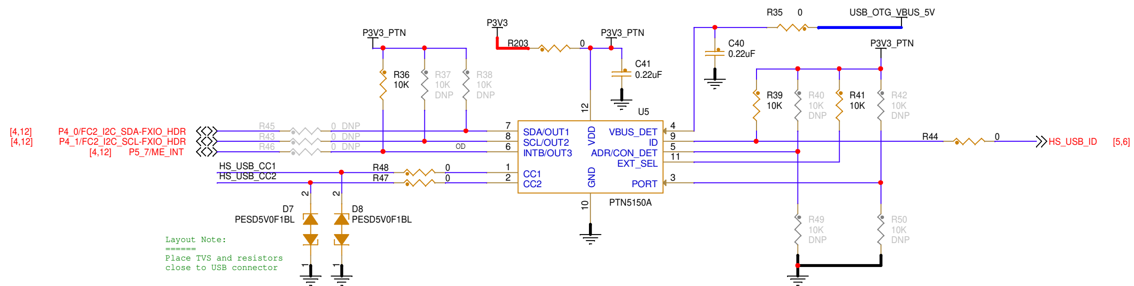
# USB2.0 Type C




# VBUS Power Control




# CC Logic



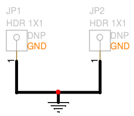

ADR/CON_DET:
When Power-up, ADR(input) function:
ADR=1: I2C Address: 0x7A(ADR)
ADR=0: I2C Address: 0x3A(ADR)
ADR=MID/FLOATING
After TINPUTLATCH, CON_DET(output) function:
CON_DET=1: Connection Detected
CON_DET=0: No Connection

EXT_SEL: External selection
High = CC1 orientation or no valid CC1/CC2 detection
Low = CC2 orientation

PORT=:
1: DFP mode
0: UFP mode
Floating: DRP mode

# MOUNTING HOLE


# Test Points




<Core Design>

Classification: Company Internal/Proprietary

Drawing Title: FRDM-MCXN947

Page Title: USB & PWR

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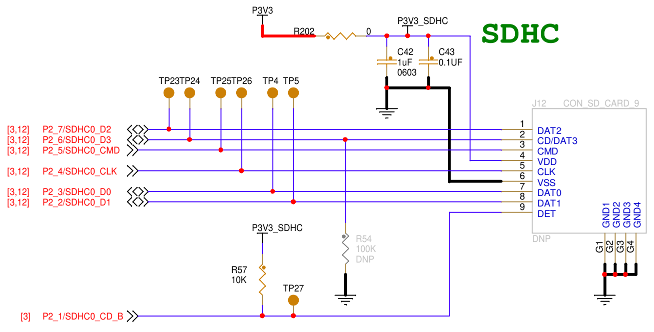
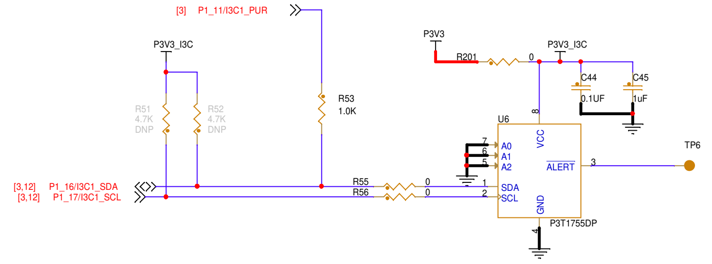
## SDHC



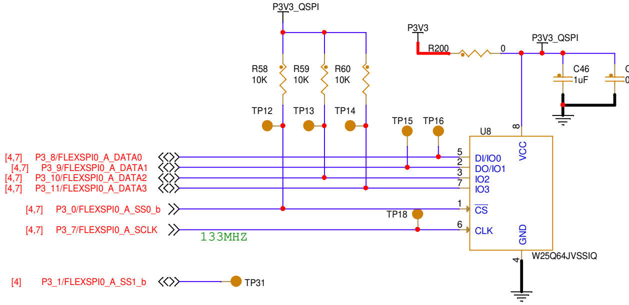

Layout Notes:
======
1) SDHC signals: equal length routing with 50ohm resistance. And as short as possible.
2) Place R54 to easily rework.

## I3C SENSOR



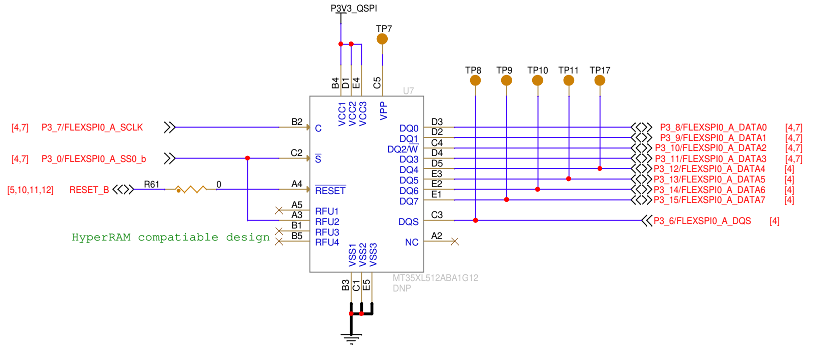

I2C ADDR: 1001000

## QSPI




Other QSPI Flash Option: MT25QL128ABA1ESE-0SIT (MICRON)

## Octal Flash




**Layout Notes:**
**======**
**1.U7 & U8 footprint overlapped.**
**2.Equal length routing with 50ohm resistance.**
**3.Place TP31 close to TP12.**

<Core Design>

Classification: Company Internal/Proprietary
Drawing Title: FRDM-MCXN947
Page Title: SD & QSPI & SENSOR
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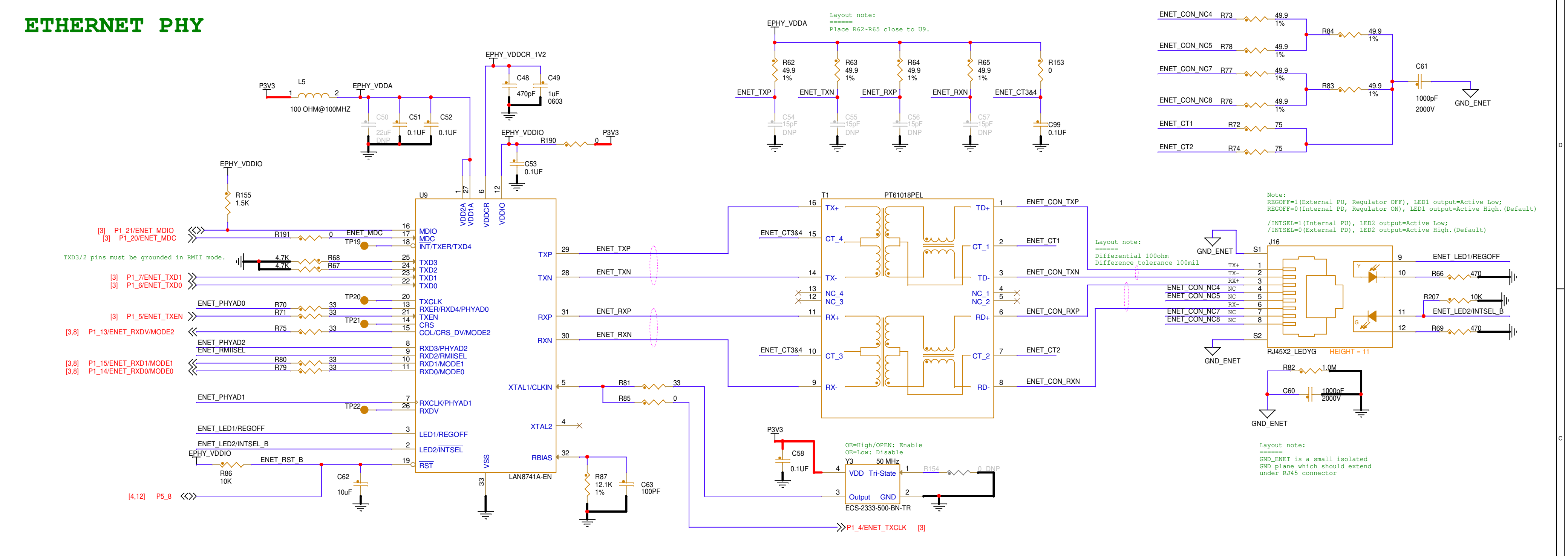
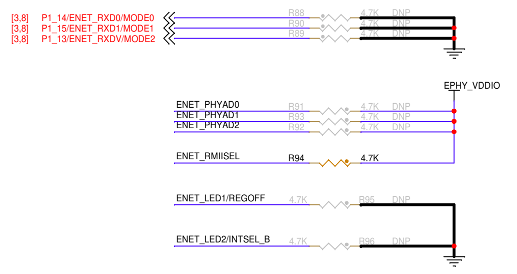
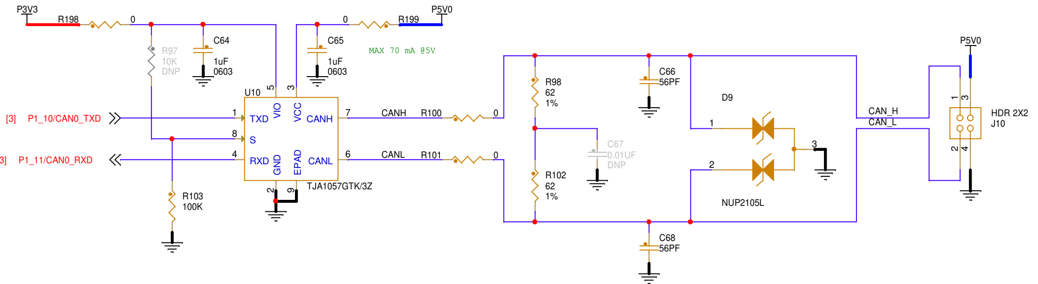
# ETHERNET PHY



| Config Pin (Internal State) | Description | Jumper Setting |
|---|---|---|
| MODE[0:2] (PU) | Mode Config<br>111 => All Capable, auto-neg enabled (Default) | Open (Int PU) |
| PHYAD[0:2] (PD) | Device Address<br>000 => Default | Open (Int PU) |
| RMIISEL (PD) | 1 => RMII (Default)<br>0=> MII | Closed (Ext PU) |
| REGOFF (PD) | 1 => Internal Regulator OFF<br>0/Floating=> Internal Regulator ON (Default) | Open (Int PU) |
| INTSEL_B (PU) | Floating 1 => nINT function (Default)<br>0=> TXD4/TXER function | Open (Int PU) |

# CAN PHY




Classification: Company Internal/Proprietary

Drawing Title: FRDM-MCXN947

Page Title: CAN PHY & ETHERNET PHY

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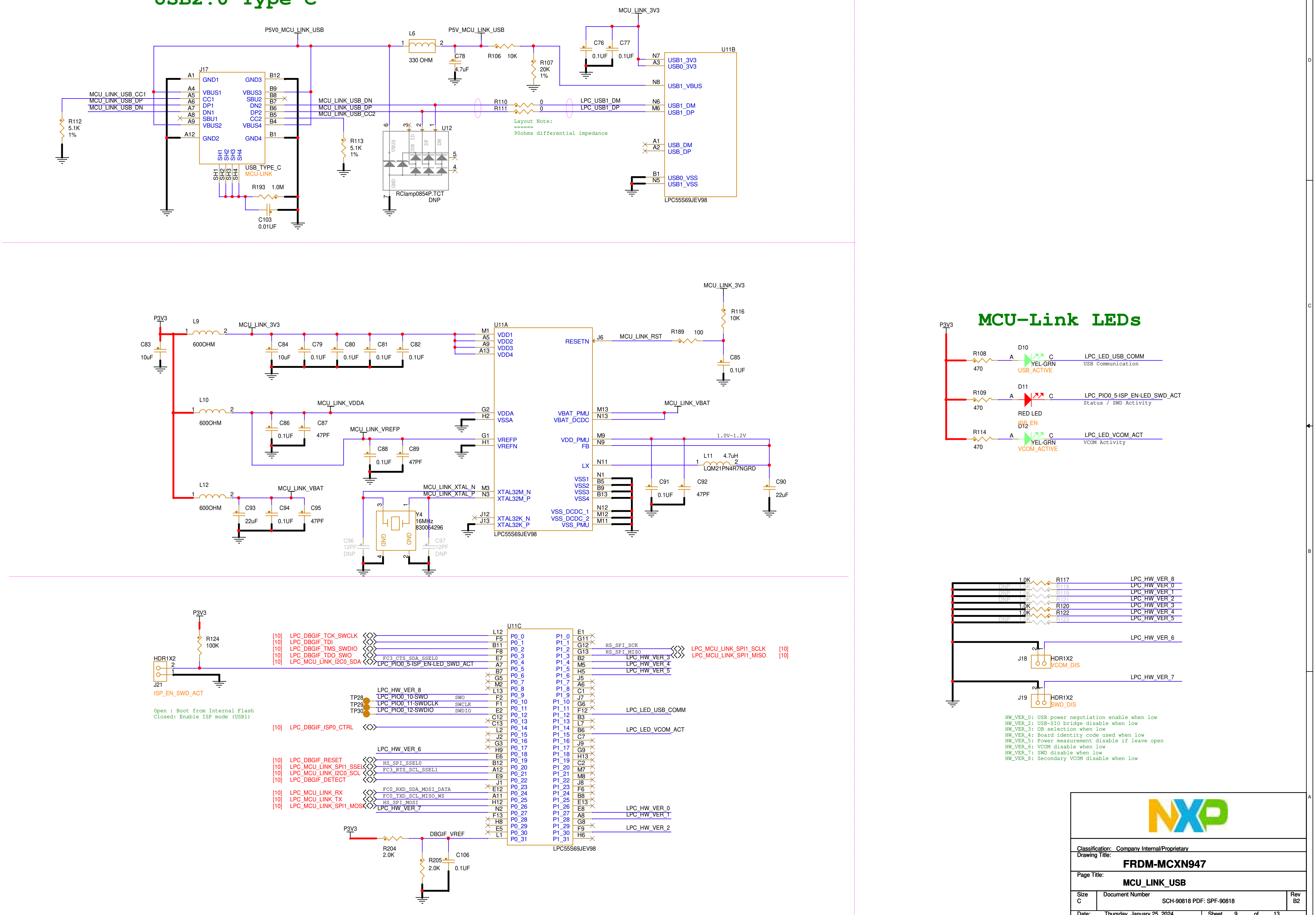
# MCU-Link USB2.0 Type C

Layout Note:
======
Add a outline for MCU LINK related components.




# MCU-Link LEDs

HW_VER_0: USB power negotiation enable when low
HW_VER_2: USB-SIO bridge disable when low
HW_VER_3: OB selection when low
HW_VER_4: Board identity code used when low
HW_VER_5: Power measurement disable if leave open
HW_VER_6: VCOM disable when low
HW_VER_7: SWD disable when low
HW_VER_8: Secondary VCOM disable when low

Classification: Company Internal/Proprietary
Drawing Title: FRDM-MCXN947
Page Title: MCU_LINK_USB

| Size | Document Number | Rev |
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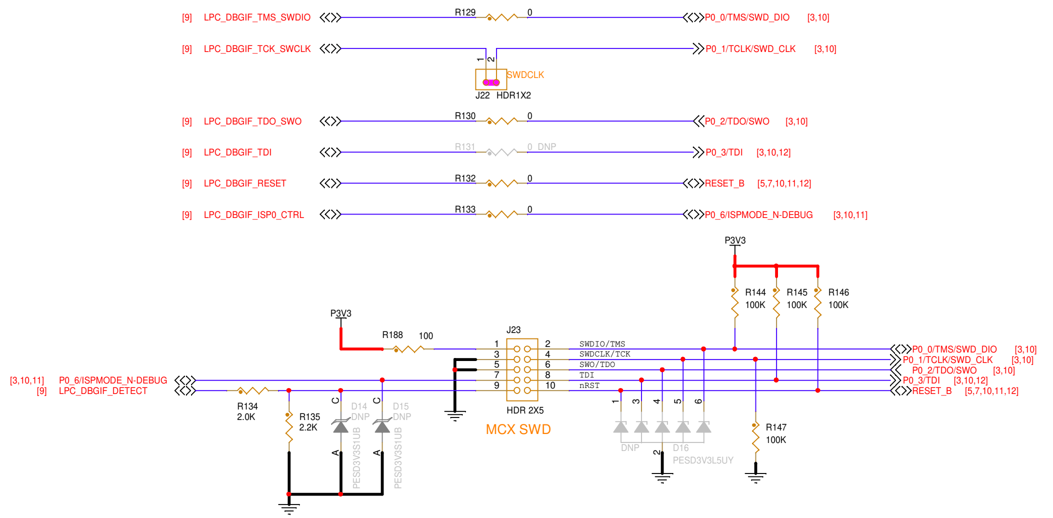
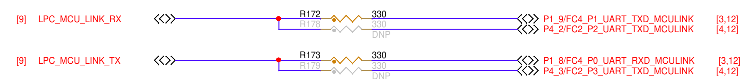
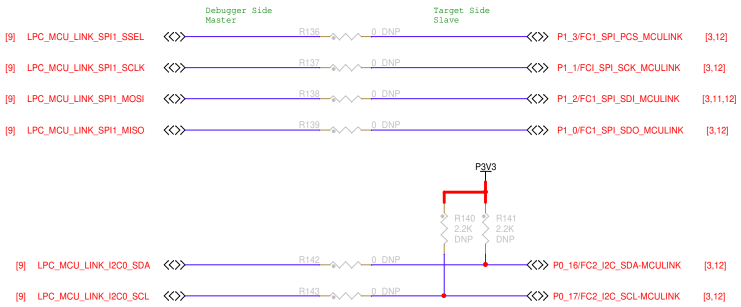
# MCU-Link Debug Interface




# MCU-Link UART




# USB Bridge




Classification: Company Internal/Proprietary

Drawing Title: FRDM-MCXN947

Page Title: MCU_LINK_DEBUG

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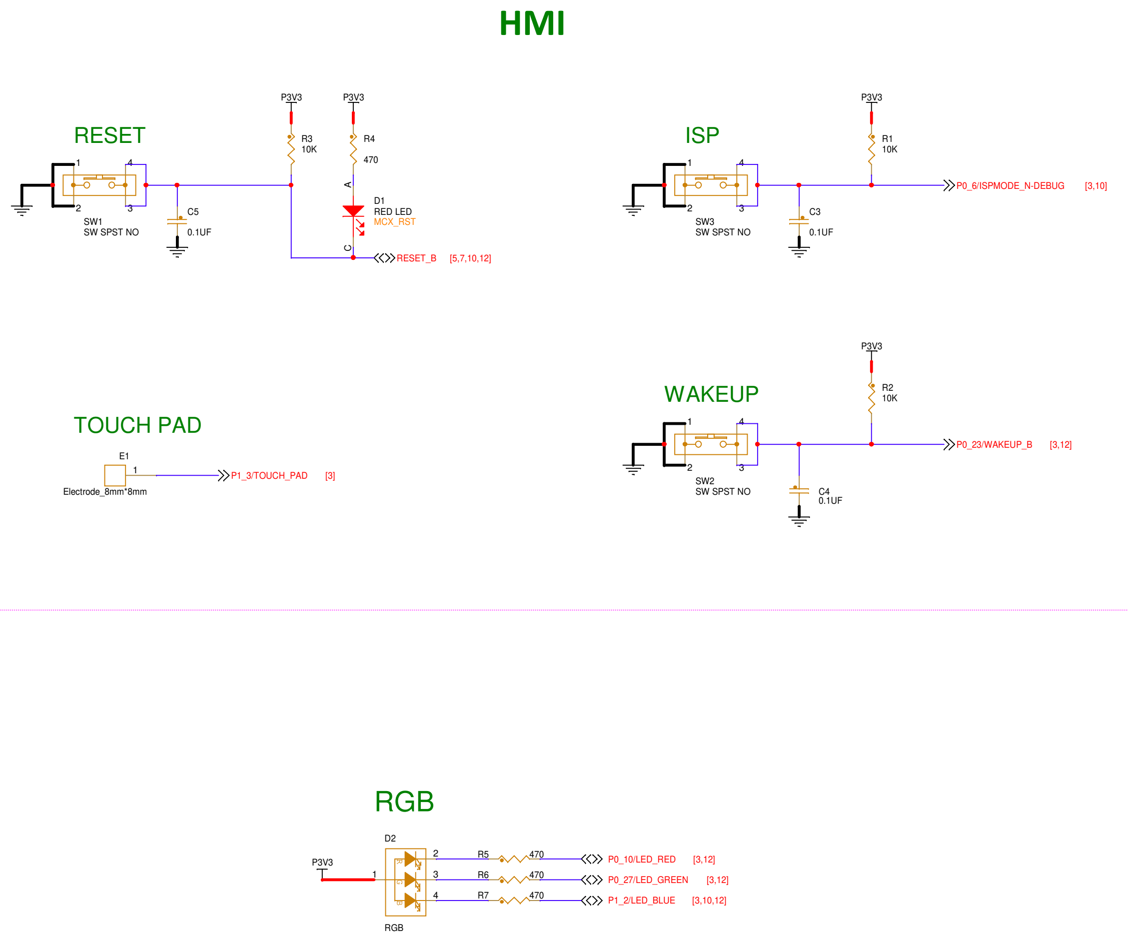


Classification: Company Internal/Proprietary

Drawing Title: **FRDM-MCXN947**

Page Title: **BLOCK DIAGRAM**

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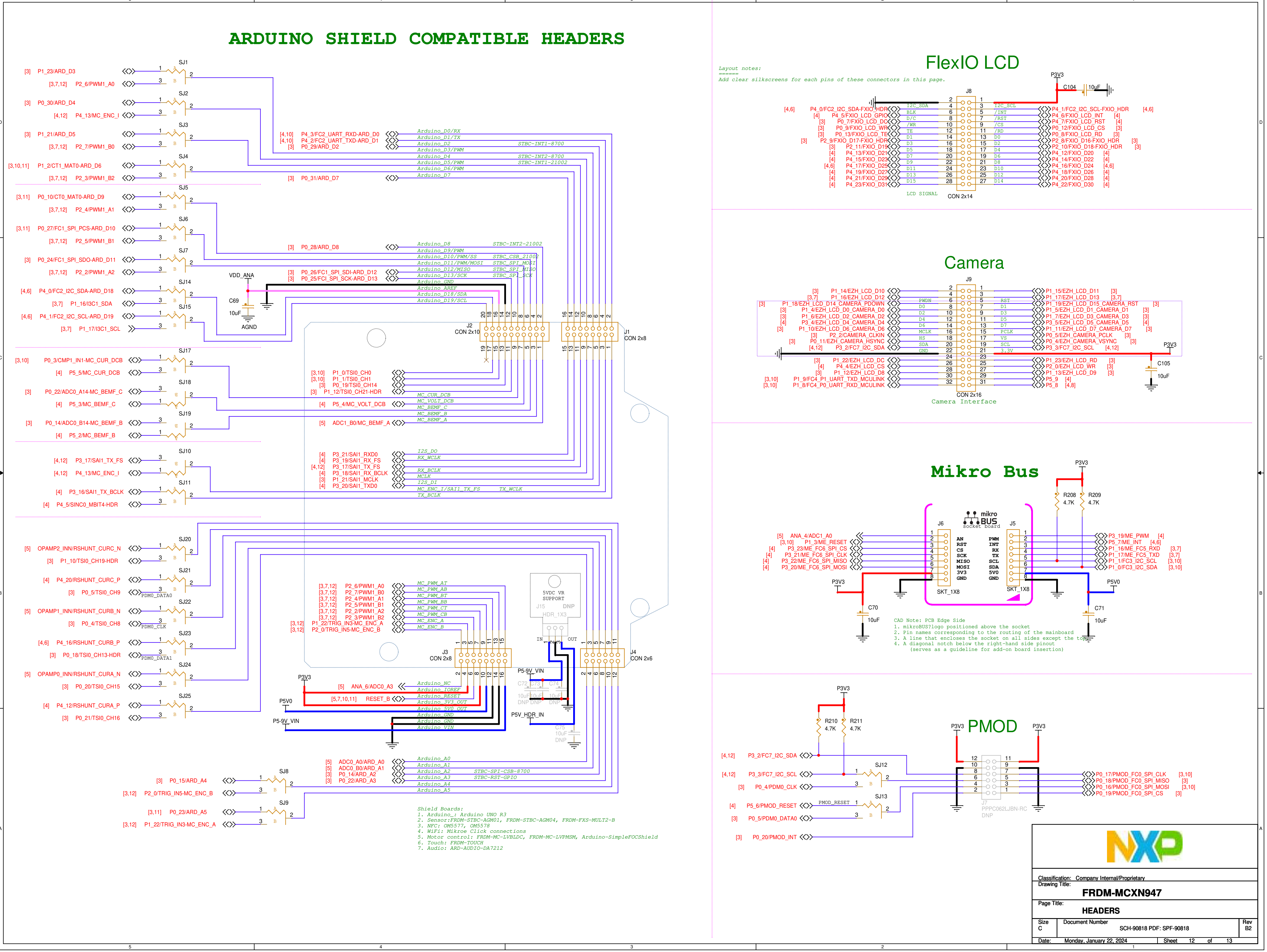

ARDUINO SHIELD COMPATIBLE HEADERS
FlexIO LCD
Layout notes:
Add clear silkscreens for each pins of these connectors in this page.
LCD SIGNAL
CON 2x14
Camera
Camera Interface
CON 2x16
Mikro Bus
CAD Note: PCB Edge Side
1. mikroBUS?logo positioned above the socket
2. Pin names corresponding to the routing of the mainboard
3. A line that encloses the socket on all sides except the top
4. A diagonal notch below the right-hand side pinout
(serves as a guideline for add-on board insertion)
PMOD
5VDC VR SUPPORT
Shield Boards:
1. Arduino_: Arduino UNO R3
2. Sensor:FRDM-STBC-AGM01, FRDM-STBC-AGM04, FRDM-FXS-MULT2-B
3. NFC: OM5577, OM5578
4. WiFi: Mikroe Click connections
5. Motor control: FRDM-MC-LVBLDC, FRDM-MC-LVPMSM, Arduino-SimpleFOCShield
6. Touch: FRDM-TOUCH
7. Audio: ARD-AUDIO-DA7212
Classification: Company Internal/Proprietary
Drawing Title: FRDM-MCXN947
Page Title: HEADERS
Size C
Document Number SCH-90818 PDF: SPF-90818
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# APPENDIX JUMPER/DNP

| REF DES | JUMPER(DEFAULT) | PAGE NAME |
|---|---|---|
| J24 | 1-2 | 06 USB & PWR |
| J18,J19 | OPEN | 09 MCU_LINK_USB |
| J22 | 1-2 | 10 MCU_LINK_DEBUG |

| REF DES | ASSY_OPT | PAGE NAME |
|---|---|---|
| J25,J26,J27,J28,JP6,R17,R20, R23,R194 | DNP | 05 SOC_PWR |
| D6,JP1,JP2,R28,R34,R37,R38, R40,R42,R43,R45,R46,R49,R50 | DNP | 06 USB & PWR |
| J12,R51,R52,R54,U7 | DNP | 07 SD & QSPI & SENSOR |
| C50,C54,C55,C56,C57,C67,R88, R89,R90,R91,R92,R93,R95,R96, R97,R154 | DNP | 08 CAN PHY & ETHERNET PHY |
| C96,C97,R118,R119,R121,R123, U12 | DNP | 09 MCU_LINK_USB |
| D14,D15,D16,R131,R136,R137, R138,R139,R140,R141,R142, R143,R178,R179 | DNP | 10 MCU_LINK_DEBUG |
| C72,C73,C74,C75,J7,J15 | DNP | 12 HEADERS |

| REF DES | SHORT(DEFAULT) | PAGE NAME |
|---|---|---|
| SJ16,SJ26,SJ27 | 1-2 | 03 SOC_PORT0~2 |
| SJ1,SJ2,SJ3,SJ4,SJ5,SJ6,SJ7, SJ8,SJ9,SJ10,SJ11,SJ12,SJ13, SJ14,SJ15,SJ17,SJ18,SJ19, SJ20,SJ21,SJ22,SJ23,SJ24,SJ25 | 1-2 | 12 HEADERS |